

Team Number 10

Project Title: Intelligent and Collaborative Autonomous Robots for Real-Time Applications

Report Date: 9/2/2024

Part one: Progress status as a team

All team members have detailed tasks listed on Teams Planner: YES

Please answer the following questions:

1. Have you met as a team this past week? If yes, give date/time, and the members attended the meeting. If no, explain.
 - a. Yes, team was able to meet in lab and over teams
 - i. All members present in lab 8/15 at 5pm
2. Have you met with the sponsor as a team? If yes, give date/time, and the member attended the meeting. If no, explain.
 - a. Yes we have met with the team sponsor.
 - i. All members present on 8/13 for Teams call
 - ii. David and Ian present for meeting 8/27 - John out of the country but still in communication
3. Describe verbally the tasks completed the past week and the challenges faced.
 - a. Worked on the presentation as well as went over the core concepts of this project. Did more research on go1 as well as yolo and object detection. Problem with restart process with cameras after once it is killed or video data.
 - b. Microsoft teams was set up as the main form of communication, and structure for the team moving forward. One of the main challenges faced is the fact that our project is more system based and software based so a lot of the tasks are likely to overlap
4. Describe the tasks to be completed the coming week.
 - a. Work on more intricate designs with the robot, such as working on making sure the cameras are working with YOLO correctly and start working on object detection with the robot. Also must find way to restart process on cameras after once it is killed for video data.
 - b. The group will be generating a live feed of footage taken from the dog in order to apply our deep learning model on
 - c. Research will be done on the onboard features of Go1, and their significance/usefulness to the project

Copy the following table from the last week's report (if this is the first report, skip):

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
David Uzan	2	0	2
Ian Simmons	2	0	2
John Sinfuego	1	0	2

Fill in the following table for this week:

	# Tasks completed past week	# Tasks not completed	# Tasks for next week
David Uzan	2	0	2
Ian Simmons			
John Sinfuego			

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Part 2: Individual Progress Report (each member complete one form)

Name of the Member: David Uzan

Report Date: 9/2/24

List the following:

1. All tasks completed the week before the past week (completion date in parenthesis). This is copied from the last report, and if this report is the first one, skip.
2. All tasks completed the past week (completion date in parenthesis)
3. All tasks you are currently working on or planned for this coming week (completion date in parenthesis)

For **each** task you have completed **during the before the past week**:

1. Task number/name

General Task 1 (8-30-24) – I met the other members of the group and project sponsor in lab 408B, where we went over basic robot operations like connecting to different modules and controlling it. We also went over the scope of the project and a road map to complete the tasks at hand.

Personal Week 1 (8-30-24) – I adapted code from Ashan's presentation of object detection on cars, which was purely identifying them as they walked into frame, to recreate a base case that instead tracks movement of the objects on an example provided at Ultralytics's github page for its YOLO modules.

For **each** task you have completed **during the past week**:

1. Task number/name

General Task 2 (9-6-24) – Implement code for video recording on dog and use it to collect video data from the Uni-Go1.

Personal Week 2 (9-6-24) – Using video recorded by the Uni-Go1, I will apply the detection algorithms to it to find exactly what the robot detects with its cameras.

2. Short description – these descriptions form the basis of the final project report
 - a. Implementation, testing, results

We used python code to access and retrieve camera feed from the dog for a set time and refresh rate and compile the video to mp4. This was then run in the same Google Colab as last week to record the results of the output using the nano model that was

used as a base case. The results were everywhere, detected things in the background as planes and trains when there were not there, but had accurate readings for people.

b. Brief and to point

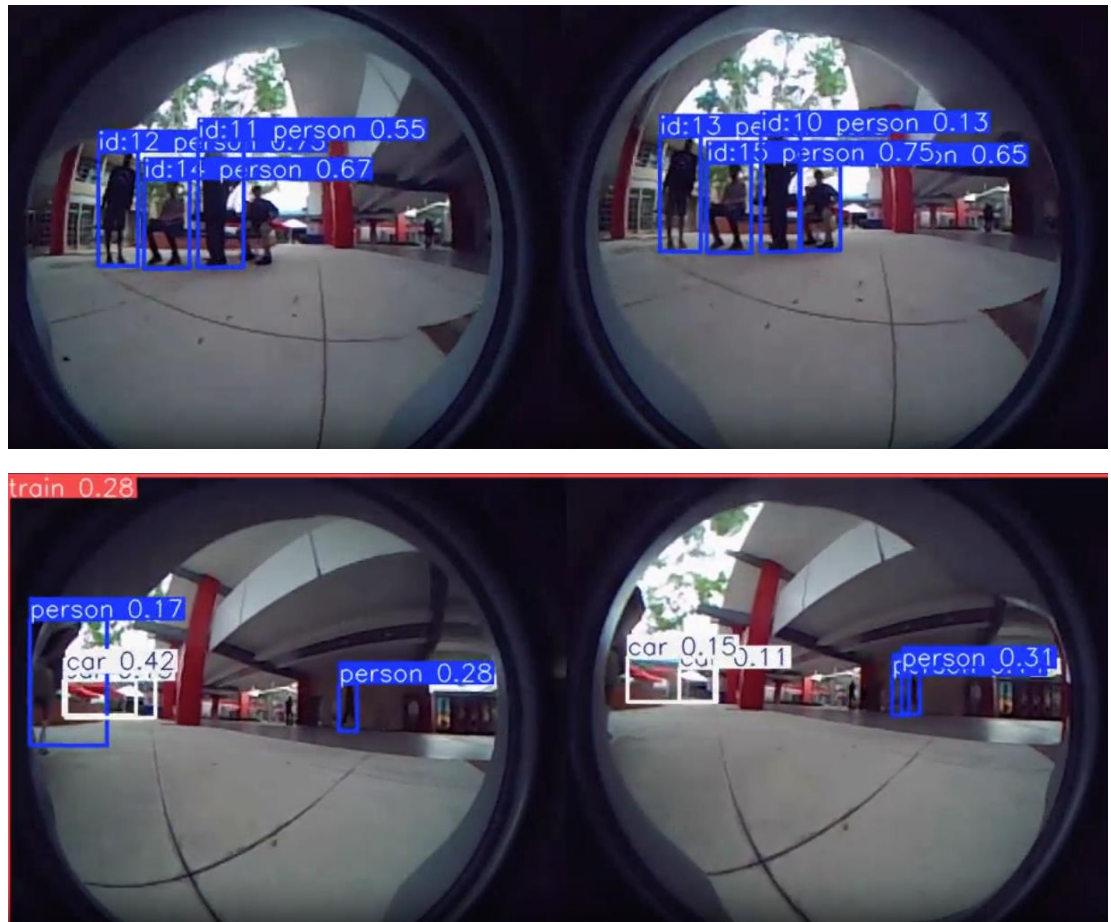
The nano YOLO module was applied to video recorded by the Uni-Go1 in the FAU breezeway and identified objects with varying certainty.

1. Outcome of the task activity

a. How did the task advance your project?

The task advanced the project greatly in our way of how the Uni-Go1 sees its environment, especially the fact that two cameras are used in the front acting as eyes.

b. Include screen shots, link to videos, plots, data tables, designs, figures, etc. as evidence of your work on the task



For **each** task you are currently working on or plan to work on in the next period:

1. Task number/name

General Task 3 (9-9-24) – Establish baseline for use cases and testing.

Personal Week 3 (9-9-24) – Migrate object detection from Google Colab to local IDE and test live feedback capabilities of the Uni-Go1

2. Short description

a. Plans, resources needed, implementation, results

The YOLO library has easy access to live video from a chosen camera by setting input equal to 0. Using this as the main driver for the experiment we will gauge the internal temperatures and wattage measured by the software provided with the Uni-Go1.

b. Main challenges, reasons for any delay in completion

Running the object detection algorithm directly on the Uni-Go1 may be hazardous for battery health and storage capacity. Offloading the information is paramount and the means of implementing that are still up in the air.

3. Outcome of the task activity

a. Include screen shots, link to videos, plots, data tables, designs, figures, etc. as evidence of your work on the task

N/A, have not started currently.